Ahmed Hannan

AI / Machine Learning Engineer | Generative AI & Computer Vision

ahmedhanan4@gmail.com • +92 333 149 5251 • linkedin.com/in/ahmedhannan2 • github.com/ahmed-hannan • Islamabad, Pakistan

Professional Summary

AI/Machine Learning Engineer with hands-on production experience designing, training, and deploying deep learning pipelines, Generative AI architectures, and computer vision systems. Proven expertise in Multimodal RAG (CLIP, FAISS, Gemma 3), cross-encoder semantic search (BERT, Amazon ESCI), real-time pose tracking (YOLO, Unity VR), and GPU compute acceleration (CUDA, Tensor Cores). Combines rigorous AI research experience at the National Center of AI (NCAI) with production full-stack engineering at EmpowerBits (awarded Performance Certificate) building high-concurrency client solutions with sub-second latency.

Technical Skills

Core AI & Deep Learning: PyTorch, TensorFlow/Keras, HuggingFace Transformers, Sentence-Transformers, Scikit-Learn
Generative AI & LLMs: Multimodal RAG, Prompt Engineering (Zero/Few-Shot, CoT), Fine-Tuning (BERT, Gemma), FAISS, spaCy, NLTK
Computer Vision & Edge AI: CLIP, YOLO (v8/v11), OpenCV, Real-Time Pose Estimation, Image Segmentation, Unity VR
HPC & GPU Optimization: CUDA, Tensor Cores, GPU Parallelization, OpenMP, MPI, Layer Offloading, FP16 Quantization
Languages & Frameworks: Python, C/C++, SQL, TypeScript, JavaScript, FastAPI, Flask, Django, PostgreSQL, MongoDB, BigQuery
MLOps & Cloud DevOps: Linux, Git/GitHub, Docker, Streamlit, CI/CD, Jira, Selenium, Apache Airflow

Professional Experience

EmpowerBits, Full-Stack Developer 2026/06 – Present

- Developed and deployed production-ready web and mobile applications (iOS/Android) serving active concurrent user traffic.
- Architected low-latency API services and reactive state management, reducing data fetch times by 30% under heavy concurrency.
- Awarded Certificate of Excellence / Performance Certificate for outstanding software delivery, technical ownership, and rapid release velocity.

RedCoast, Full-Stack Software Engineer 2025/06 – 2026/06

- Contributed to product optimization and feature delivery across multiple client-facing production applications.
- Automated regression testing workflows with Selenium and Jira; analyzed large-scale event logs with Google BigQuery to optimize system bottlenecks.

National Center of AI (NCAI, NUST), AI Research Intern 2024/08 – 2024/09

- Researched generative transformer architectures and multimodal image-text synthesis; authored formal technical documentation for hybrid AI deployments.

FAST – NUCES, Teaching Assistant (Programming Fundamentals) 2024/06 – 2024/09

- Mentored 50+ students in C/C++ memory management, dynamic data structures, and algorithmic problem-solving.

Featured AI/ML Projects

Pose2Play — AI-Powered VR Rehabilitation Platform (Final Year Project) GitHub: Pose2Play

- Built a real-time computer vision system using YOLO-based pose estimation to track 3D patient movement kinematics and deliver corrective feedback.
- Integrated low-latency Python model inference with Unity VR, achieving 30+ FPS real-time responsiveness to boost physical therapy adherence.

Multimodal RAG & Semantic Product Search Suite GitHub: multimodal-rag-semantic-search

- Engineered an end-to-end multimodal RAG pipeline for complex financial PDFs using PyMuPDF, OCR, and dual encoders (CLIP-512 + MiniLM-384) in FAISS.
- Deployed quantized Gemma-3-1b-it inference with FP16 and layer offloading for 3GB VRAM; fine-tuned BERT-Tiny cross-encoder on Amazon ESCI (1.1M records).

CRAG: Advanced Multi-Pipeline RAG Benchmark GitHub: crag-advanced-rag-pipelines

- Benchmarked 7 retrieval pipelines (RAG Fusion, HyDE, Corrective RAG with confidence gating, Graph RAG, RRR) on the Meta CRAG noisy web corpus.
- Developed a global FAISS vector index, spaCy NER answer extractor, and evaluation harness reporting normalized exact-match accuracy.

High-Performance Deep Neural Network Acceleration GitHub: Neural-Network-Acceleration

- Accelerated deep neural network forward/backward passes and matrix kernels using CUDA, Tensor Cores, and GPU parallelization techniques.

Education

National University of Computer & Emerging Sciences (FAST-NUCES), BS in Computer Science Graduation: June 2026

- Relevant Coursework: Deep Learning, Generative AI, Machine Learning, Computer Vision, High-Performance Computing, Algorithms, Linear Algebra.